GE Medical Systems

Technical Publication

Direction 2270001-100
Revision 0

GE Medical Systems
Retrospective/Prospective Gating QX/i

LEGAL NOTES

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。

ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

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注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触
电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write “Damage In Shipment” on ALL copies of the freight or express bill BEFORE delivery is accepted or “signed for” by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont

Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

NOTES:

Revision History

Revision	Date	Reason for change
0	June16, 2000	Initial product release

List of Effected Pages

PAGES	REVISION	PAGES	REVISION	PAGES	REVISION
1 through 54	0				

Preface
Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by it format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task.

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0
Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted form GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.

CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

DANGER
EXCESSIVE VOLTAGE
CRUSH POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING ROTATING EQUIPMENT BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH <u>COULD</u> OR CAN <u>CAUSE SERIOUS</u> PERSONAL <u>INJURY</u> OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

NOTICE Equipment Damage Possible

Notice is used when a hazard is present that can cause property damage but has absolutely <u>no personal injury risk</u>.They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

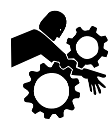
It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

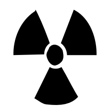
Important information will always be preceded by the exclamation point contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL

MECHANICAL

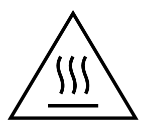


RADIATION

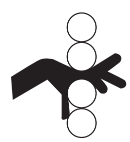


LASER

HEAT

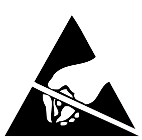


PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT

WEAR EYE PROTECTION

Section 2.0
Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

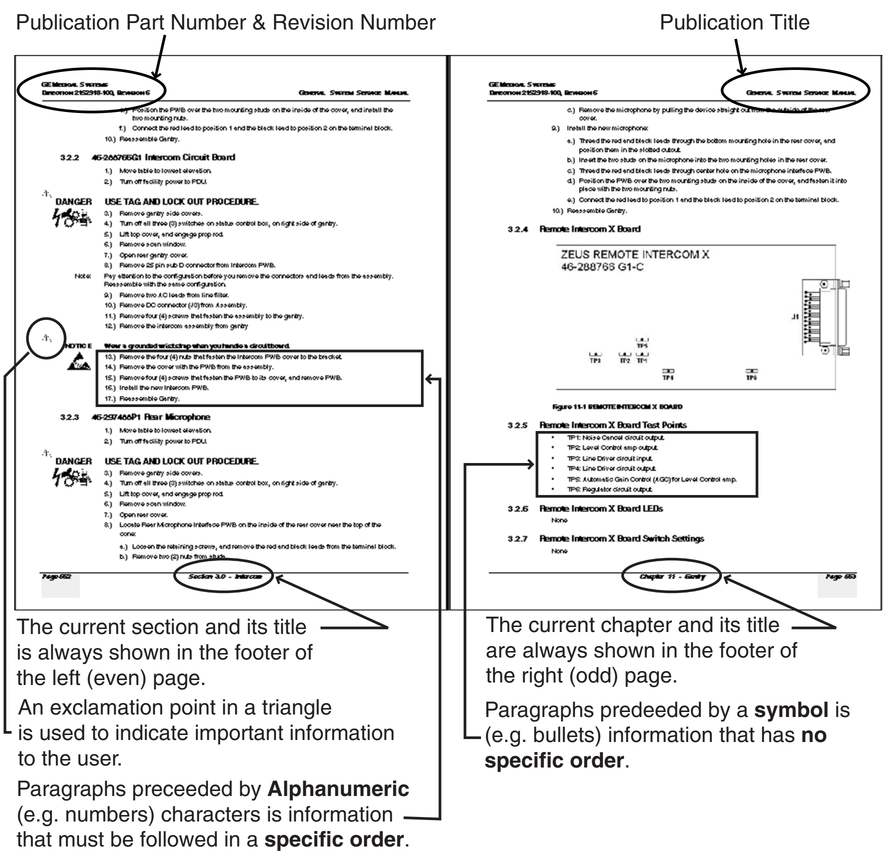
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: **Conveys information that should be considered important to the reader.**

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directionality. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output

This paragraph denotes computer screen fixed output. It’s output is fixed from the sense that it does not vary from application to application.It’s the most commonly used style used to indicate filenames, paths and text.

Example: Variable Output

This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>

Example: Fixed Input

This paragraph denotes fixed input. It’s typed input that will not vary from application to application. Fixed text the user is required to supply as input.

Example: Variable Input

This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.

End of Section

Table of Contents

Chapter 1
Introduction

Follow the instructions in this document to upgrade the HSA,CT/i, or LightSpeed gantry to accommodate an EKG machine for use during cardiac studies.

Section 1.0
Cardiac Calcification Theory

1.1 Retrospective Gating

1.1.1 Operational:

During the cardiac study, the technologist attaches EKG monitor leads to the patient, and acquires a Cine study of the chest area. Upon completion of the study, the Advantage Windows Cardiac software uses the EKG data to extract images from the study that show the heart at rest. Physicians can examine the motion free heart images for signs of calcification, or other pathology.

1.1.2 Hardware - Control of EKG data recording:

For Retrospective Gating the XRAYCOM signal found on STC connector A1 A21 is fed through the Driver to the Opto Isolator, Pins W2, W3, W4 and W5. From there it is sent to the EKG device to enable EKG waveform storage on a floppy disk during the XRAY ON period. Detailed information about this recording activity is defined in the EKG instruction manual.

1.2 Prospective Gating

1.2.1 Operational:

1.2.1.1 Mode 1 (Scout)

This mode functions as a normal Scout with the exception that there is a selection button in the Scout View/Editor that allows the selection of "Gating". When this button is selected (manually or via protocol), the button becomes backlit with blue and the gating hardware and firmware is enabled. With proper connections and an EKG signal provided to the system, the heart rate will be displayed in the button.

If the hardware has a fault or is not connected properly, the button will become red and the Confirm button will not be active. In addition, a message window will appear indicating a problem has been detected.

An additional error condition can happen if the heart rate input stops for a duration of greater than 5 beats. This condition will result in an error message window appearing. The heart rate must be re-established and the error acknowledged. The delay in this error message is intended to minimize false errors due to irregular heart rates.

1.2.1.2 Mode 2 (Cine)

The image reconstruction of a heart requires data from a minimum of 240 degrees in a single plane of rotation. For the data to produce optimum images, the heart must be scanned during a time of no or little motion. The Gating device allows us to know when the heart will be in a state of rest. With this information we can program the scanner to take images during that rest cycle. If the heart rate is too high, motion artifact will appear in the images.

The parameters that we are dealing with are:

TRIGGER DELAY - Time from receiving gating pulse until X-ray is turned on.

This value is set in the Scan RX and can be from 30% to 60% of the cardiac cycle.The default setting is 47%.

INTERVAL - Time from image to image

This value is set in the Scan RX and can be set to 50 or 100 milliseconds.

NUMBER OF IMAGES - The total number of images selected for a single cardiac cycle.

This value is set in the Scan RX and can be set to 1, 3, or 5

DURATION - Time of X-ray on

This value is a calculated value based on the Scan RX values listed above. The equation for Duration is:

Duration = (Number of Images - 1) (Interval) + 530 milliseconds

1.2.2 Hardware:

1.2.2.1 General

The 2252435 Driver is installed on the back of the STC. Cable 2252473 is located inside the Gantry and connects the Driver to the Isolator 2252437, which is located at the base of the Gantry. 2252472 Cable couples the EKG to the Gantry. Installation Drawing 2269789IDW, Figure 1-1, shows the location of each of these items in more detail.

1.2.2.2 Transfer of "R" pulse from EKG Device to STC

The "R" pulse from the EKG is coupled into the Isolator at Pins W6+ and W7. The 10 ms wide, heart rate signal called R-Pulse, is buffered and sent along to the to the Driver board. An LED flashes on the Opto-Isolator board during each heartbeat.

The R-Pulse signal is filtered and re-formatted on the Driver board before it is sent into the STC on Pin A2 B4. Another LED on the Driver board flashes with each heartbeat.

At the release of this option the Sonata inputs are not utilized. When the Sonata input is made operational, the Prospective Cardiac / Sonata functions are mutually exclusive. The selection of function to be enabled is done by the SonEna+ (A2 B16) and the SonEna- (A2 B17) signal from the STC. This selection is controlled by the firmware.

1.2.2.3 Interface to "Sonata" foot control

Signals from the standard Sonata foot control are coupled into the Driver board at J1-1, J1-2, J1-3 and J1-4. The Pedal signal appears at Option - (A2 B4) when the Sonata option is selected by signals at A2 B16 and A2 B17.

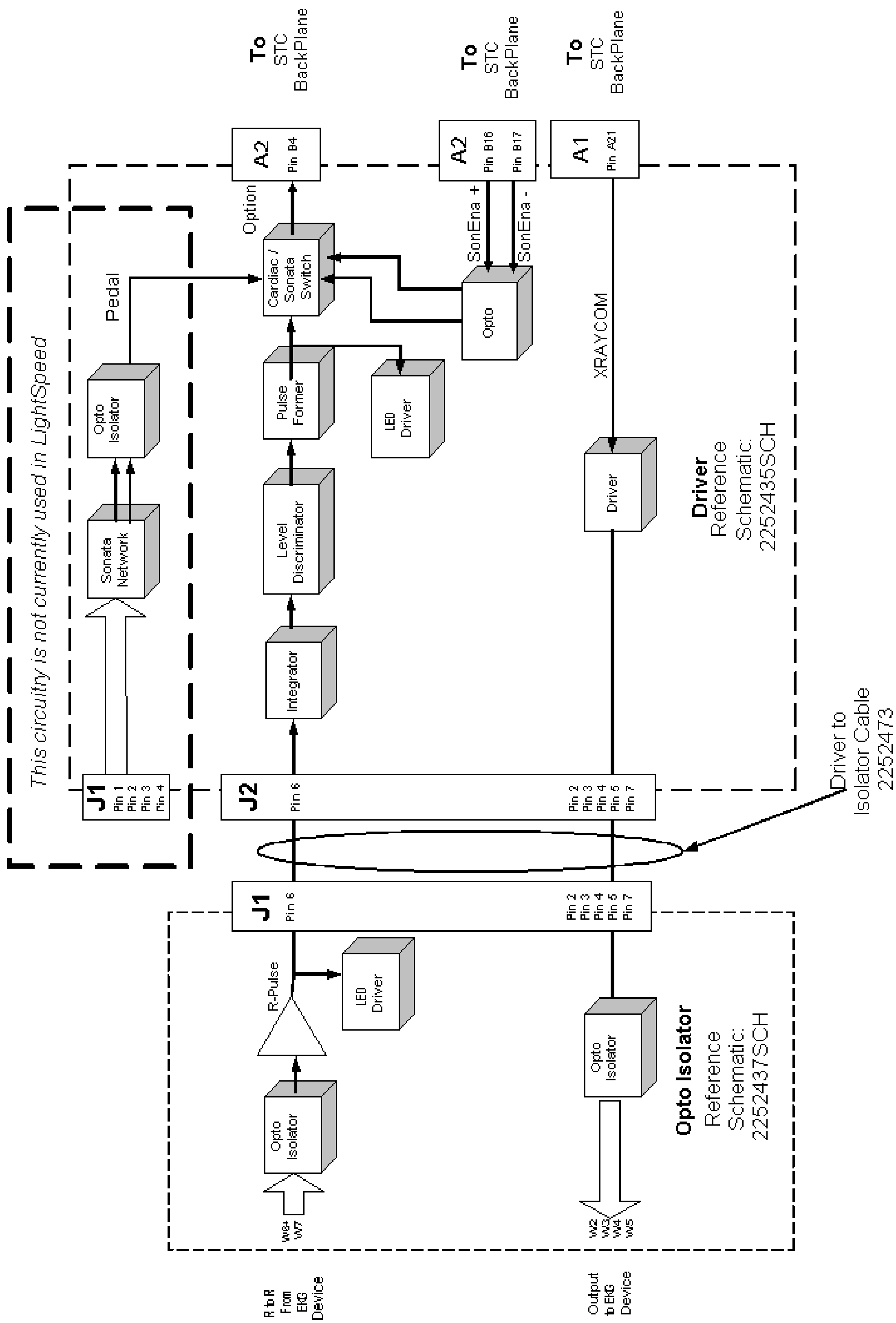


Figure 1-1 Installation Drawing

Section 2.0 Requirements

This publication assumes the facility has an Advantage Windows system with version 3.1, or higher, software.

It is also required that this option be used on a LightSpeed.

Section 3.0 Manual Description

In addition to this introductory chapter (Chapter 1), this manual contains the following:

- Chapter 2: Upgrade kit parts list and upgrade flowchart/checklist
- Chapter 3: Hardware upgrade procedures
- Chapter 4: System functional checks
- Chapter 5: Software installation information
- Chapter 6: System interconnects
- Chapter 7: Troubleshooting

Chapter 2
Upgrade Overview for SmartScore

Section 1.0
SmartScore Upgrade Parts List

✓	PART NUMBER	DESCRIPTION
	2252435	Prospective Cardiac Driver
	2252473	Prospective Cardiac Driver to Opto-isolator Cable
	2252474	Gantry to Opto-isolator Board/Cable Assembly
	516A770P405	Screw
	46-136375P32	Hex Spacer, $^3/_{16}$" hex. 0.88" long
	46-328421P14	Screw, Pan Head 4mm, 10mm
	2215053	Left Base Cover
	2216584	Cardiac Coverplate
	2218235	Cover - STC Cardiac option
	2252472	Gantry to EKG Cable
	46-302200P3	Rating Plate
	2270001-100	Retrospective/Prospective Gating Service Manual
		Option MOD
	2228113-4	SmartScore AWW CD-ROM
	E8007N • E8007MG • E8007ME	Starter Kit • Box of Floppy Disks • EKG Electrodes
	E8007MH	Heartbeat Simulator

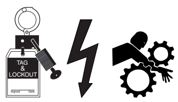
Table 2-1 SmartScore Option Parts List

Section 2.0
SmartScore Upgrade Overview

Complete the following steps to install the SmartScore upgrade. Each step listed will be discussed in more detail on the following pages.

Note: **Some SmartScore kits come with an optional disk drive for the Octane to increase the customers image space. This optional disk drive can be installed at the same time as the gantry hardware is being installed. Also, the optional disk drive could be broken out from this installation and be installed at a later date and time depending on customer time constraints. If postponed, be aware the customers image space could be limited.**

WARNING **REMOVE POWER, AND REMEMBER TO TAG AND LOCKOUT HAZARDS.**



1.) Remove gantry covers (Chapter 3).
2.) Install cardiac driver and cardiac harness (Chapter 3).
3.) Install gantry base cover (Chapter 3).
4.) Install Opto-isolator board and cable (Chapter 3).
5.) Unpack and install EKG machine (follow manufacturer's instructions).
6.) Restore system power.
7.) Install AW 3.1 software (Chapter 5).
8.) Check system function (Chapter 4).
9.) Distribute operators manual and accessories kit to customers.

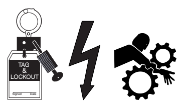
Please keep track of your progress.

Chapter 3
Hardware (Gantry) Upgrade

Section 1.0
Remove Covers from Gantry

1.) Shut system down and remove power at the Main Disconnect.

WARNING **WHEN REMOVING POWER, REMEMBER TO TAG AND LOCKOUT IT OUT.**



2.) Remove both side covers from the gantry.

3.) Turn off the following gantry service switches:
- 550VDC
- 120VAC
- Gantry Rotate (Axial Drive Enable)

4.) Open the front and rear gantry covers.

5.) Remove, and set aside, the left side cover from the gantry base.

Section 2.0
Install Cardiac Driver Board (2252435) and Harness (2252473)

1.) Remove the STC cover to expose the STC backplane and set aside (refer to Figure 3-1).

2.) Remove the screws that fasten the harness bracket to the bottom of the STC.

3.) Temporarily remove the A3-J2 and A3-J3 cables from the STC backplane.

a.) Remove the cardiac driver from the upgrade kit.

b.) Plug the cardiac cable into J2 on the cardiac driver.

c.) Orient the cardiac driver with the J2 pins pointing toward the floor, and plug it into A3-J2 and A3-J3 on the backplane.

d.) Use hardware from the kit to fasten the cardiac driver to the spacers on the STC backplane.

e.) Plug the A3-J2 and A3-J3 cables into the corresponding connectors on the cardiac driver.

f.) Use the captive hardware on A3-J2 and A3-J3 to fasten the connectors into place.

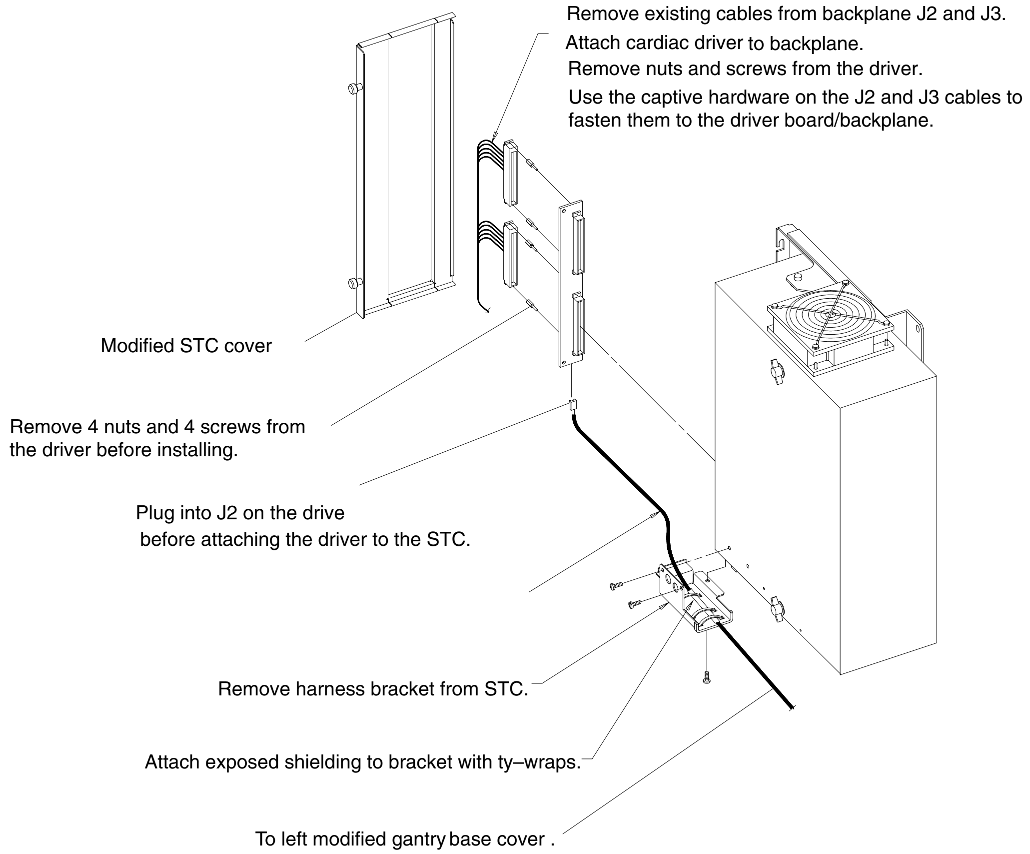


Figure 3-1 Add Cardiac Driver to STC Backplane

4.) Use ty-wraps to fasten the exposed shielding on the end of the cardiac harness to the existing cable shields.

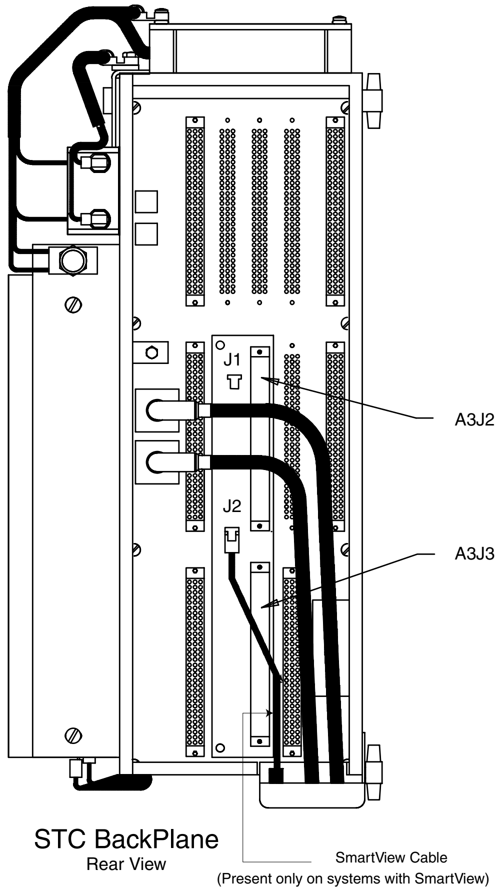


Figure 3-2 STC Backplane with the Cardiac Driver in Place

5.) Reattach the harness bracket to the bottom of the STC.

6.) Remove the modified STC backplane cover from the upgrade kit, and fasten it into place. (Discard the old STC cover.)

7.) Route the cardiac harness down the rotating base and cable router (refer to Figure 3-3).

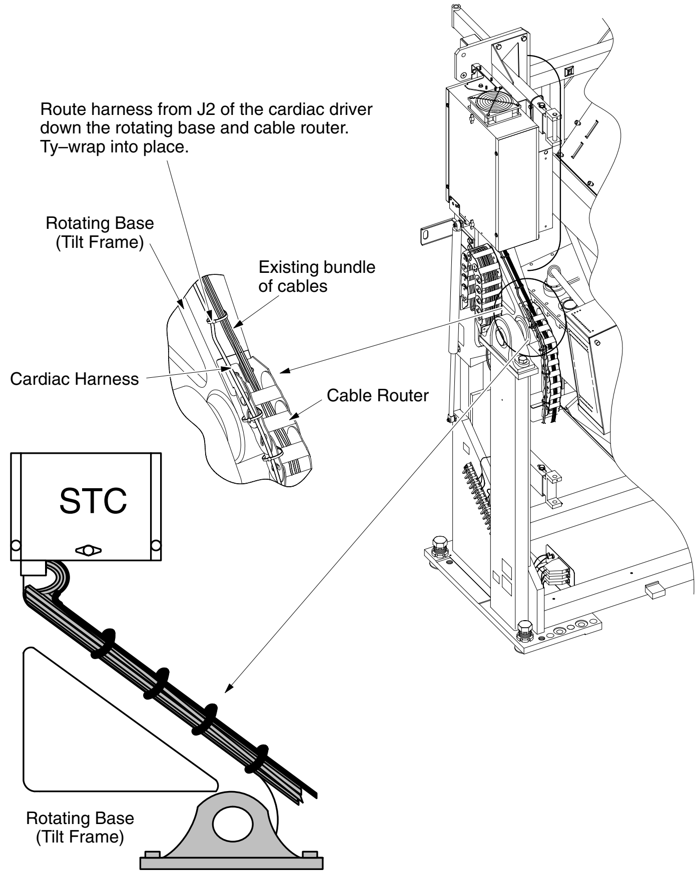


Figure 3-3 Route the Cardiac Harness Down the Rotating Base

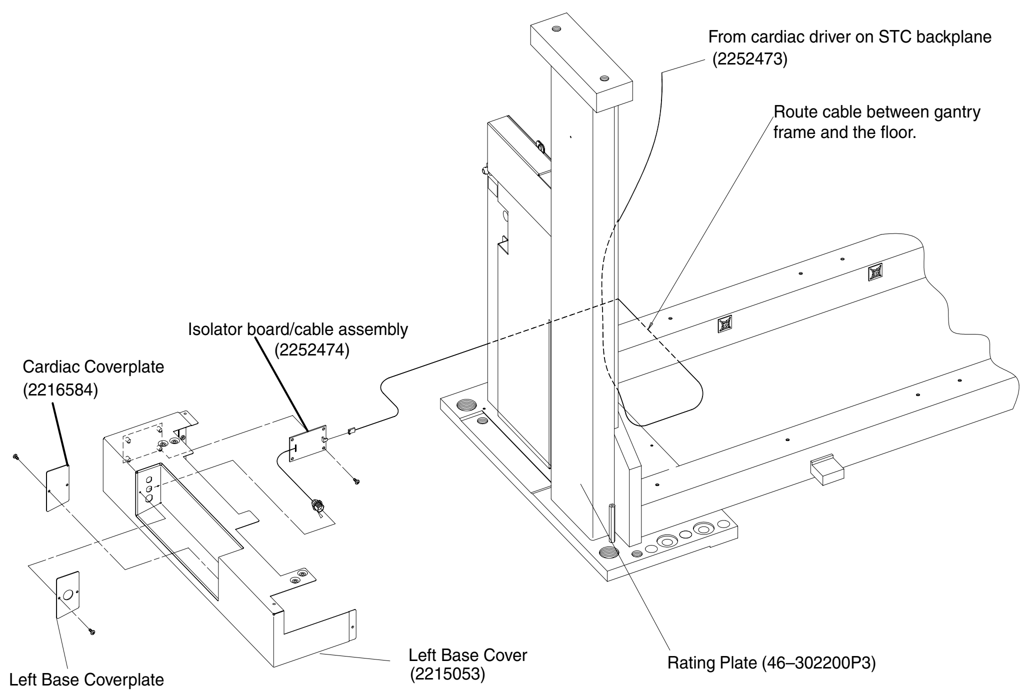


Figure 3-4 Route the Cardiac Cable In the Gantry Base

8.) Route the cable connector under the gantry base, next to the floor, to avoid interfering with the covers (refer to Figure 3-4).

Section 3.0
Install Gantry Base Cover

1.) Remove the cardiac cover plate, 2216584, and left base plate cover, 2215053, from the upgrade kit.
2.) Attach the cardiac cover plate to the left base plate cover.

Section 4.0
Install Opto-Isolator Board

1.) Attach the cable from the isolator board/cable assembly, 2252474, to the middle hole on the gantry base cover (refer to Figure 3-4). (If you attached the correct plate, you'll only see one hole.)
2.) Attach the isolator board/cable assembly board to the gantry base cover.
3.) Plug cable 2252473 into J1 of the isolator board/cable assembly 2252474.
    If cable 2216886 exists in the gantry rotating base cable router, it can be removed and discarded.
4.) After you route the cardiac cable into place in the gantry base, tie/wrap it to the existing cables.

Chapter 4
System Functional Checks

Purpose: Use this procedure to ensure the proper installation and operation of the CACS EKG monitor. Install the CACS software package on the Advantage Windows workstation before following this test procedure.

Section 1.0
Create Test EKG Waveform

1.) Turn on EKG monitor power:
   - a.) Turn ON the power switch on the front of the monitor.
   - b.) Turn ON the power switch on the back of the EKG monitor.

2.) Wait for the EKG monitor to pass its power-up self-tests.

3.) Locate the “seat-belt” GE Connect symbol in the bottom right corner of the monitor screen.
   - If necessary, disconnect the cable between the EKG monitor and gantry base.
   - Make sure the two halves of the symbol do not touch, which indicates a disconnected gantry cable.

4.) Connect the cable from the left CT gantry base to the back of the EKG monitor. Make sure the “`seat-belt`” symbol closes.

5.) Connect the gray EKG interface cable to the front of the monitor.

6.) Connect the black EKG lead cable to the interface cable by matching the colored ends of the flexible cable to the corresponding colored connectors on the EKG interface cable.

7.) Use the RA (white), RL (green), LL (red), and LA (black) connectors to attach the black flexible lead cable to the EKG heartbeat simulator .

8.) Turn on the EKG heartbeat simulator and set the heart rate to between 60 and 80 bpm (beats per minute). Select the EKG output waveform and observe the waveform on the display.

9.) Locate the disk drive on the side of the EKG monitor, and insert a formatted, blank high density floppy disk into the drive opening.

10.) On the console, prescribe and acquire a 15 second helical scan of air.

11.) Record Patient ID number:__________________________

12.) Enter the same Patient ID into the EKG machine and press enter. The status box in the bottom right corner of the EKG display turns from black to green.

13.) Select CONFIRM and acquire the scan.
   - The status box on the EKG display turns from green to yellow when x-rays begin.
   - The status box on the EKG display turns from yellow to black when the scan completes.

14.) Wait for the EKG monitor to save the EKG data to the floppy disk.

15.) After the Patient ID screen appears, eject the disk from the drive.

Section 2.0 View the Test EKG Waveform

Note: Save the floppy you create in the following procedure. The floppy is used in loading software in Chapter 5.

1.) Take the disk to the Advantage Windows workstation and insert it into the disk drive.
2.) Start the CACS application.
3.) Select the INPUT tab
4.) Select UPDATE to locate the EKG file on the floppy disk.
5.) Select the EKG file name, and click OK.
6.) Select the ANALYSIS tab. Make sure the EKG waveform appears in green at the top of the screen.
7.) Return to the INPUT tab, select EJECT, and remove the disk from the floppy drive.
8.) Quit the CACS application.

Section 3.0 Prospective Gating Functional Testing:

This section is intended to be a system level test for the functionality of the Retro/Pro - Gating Option. Three scans will be performed during these tests. First, a Scout scan will be used to verify connection and recognition of the EKG signal from the patient through to the scanner. Second, a Gated Cine scan will be completed to verify the proper operation of the scanner with the EKG signal controlling the X-ray Start signal. This test will also verify communication back to the EKG device for the recording of the EKG signal. The third and final test will verify that stopping the heart rate will cause the scanner to complete the current image and wait for the next EKG pulse before starting the next image.

3.1 Test Setup:

1.) Position a phantom on the table for imaging during the following tests.
2.) Attach an EKG simulator to the EKG device and verify proper operation of the EKG device (see operator manual for EKG device).
3.) Connect EKG device output to the Gantry per installation instructions.
4.) Select a heart rate on the EKG simulator (75 bpm [Beats Per Minute]).
5.) Select the standard SmartScore acquisition protocol (SmartScore 60 BPM Gated).

3.2 Scout Scan:

1.) Set system as detailed in Test Setup.
2.) In the Scout View/Edit screen select the "Monitor Gating" button.
3.) Verify that the heart rate seen on the EKG device matches the display in the gated button on the Scout View/Edit screen (+/- 1 beat per minute).
4.) Change the heart rate on the EKG device and verify that the scanner display changes to match that on the EKG device

Note: There will be a delay to the update on the scanner display due to 4 beat averaging done in the scanner.

5.) Turn off the EKG device and verify that the following error message is displayed in a popup window.

`Failed to detect heart rate, please check connection.`

6.) Turn the EKG device on and press <OK> in the error message window.
7.) Verify that the heart rate is once again displayed on the scanner.
8.) Complete the Scout scan and verify normal operation of the localizer for defining the scanning range.

3.3 First Image Series:

1.) Set system as detailed in Test Setup.
2.) Set the EKG simulator for a heart rate of 75 bpm.
3.) Set the trigger delay to default value.
4.) Select 1 images per R to R interval.
5.) Verify the following values:
R TO R INTERVAL= 800 M. SEC.
TRIGGER DELAY = 376 M. SEC.
CINE DURATION= 600 M. SEC.
6.) Complete the scan and confirm that recon creates images.

3.4 Second Image Series:

1.) Following the First Image Series, select "Repeat Image Series"
2.) Set the EKG simulator for a heart rate of 75 bpm.
3.) Set the trigger delay to default value.
4.) Select 1 images per R to R interval.
5.) Verify the following values:
R TO R INTERVAL= 800 M. SEC.
TRIGGER DELAY = 376 M. SEC.
CINE DURATION= 600 M. SEC.

6.) Begin the scan and turn off the EKG simulator after the first image is complete.

7.) Verify that the current image is completed and the following error message is displayed in the Real Time Information Window.

```
Scanner Hardware Stopped Scan
```

8.) Turn the EKG simulator on, select <OK> in the error message window and press <Resume>.

9.) Verify that the scan completes and recon creates images.

Chapter 5
SmartScore Option (Software) Installation

Section 1.0
SmartScore Option Software Load on AWW

The CT system does NOT require a software upgrade to accommodate the SmartScore hardware. The SmartScore option requires at least a version 3.1 Advantage Windows Workstation (AWW). Follow the instructions for AWW software installation that arrived with the AWW 3.1 software package.

1.) Load software on Advantage Windows 3.1 system. Instructions can be found in side the cd-rom case (CD-ROM case front page insert - back side) or in the `readme.txt` file on the CD-ROM.
2.) Verify that you can read the EKG waveform file saved in Chapter 4 Section 2.0. Insert the floppy and read the EKG waveform from it.

Section 2.0
SmartScore Pro Option Load on Scanner

The SmartScore Pro option requires the enabling of the system software and firmware that is utilized for this option. This is done via an option key. Please refer to the LightSpeed Installation Manual for option installation instructions.

Chapter 6
SmartScore (System) Interconnect

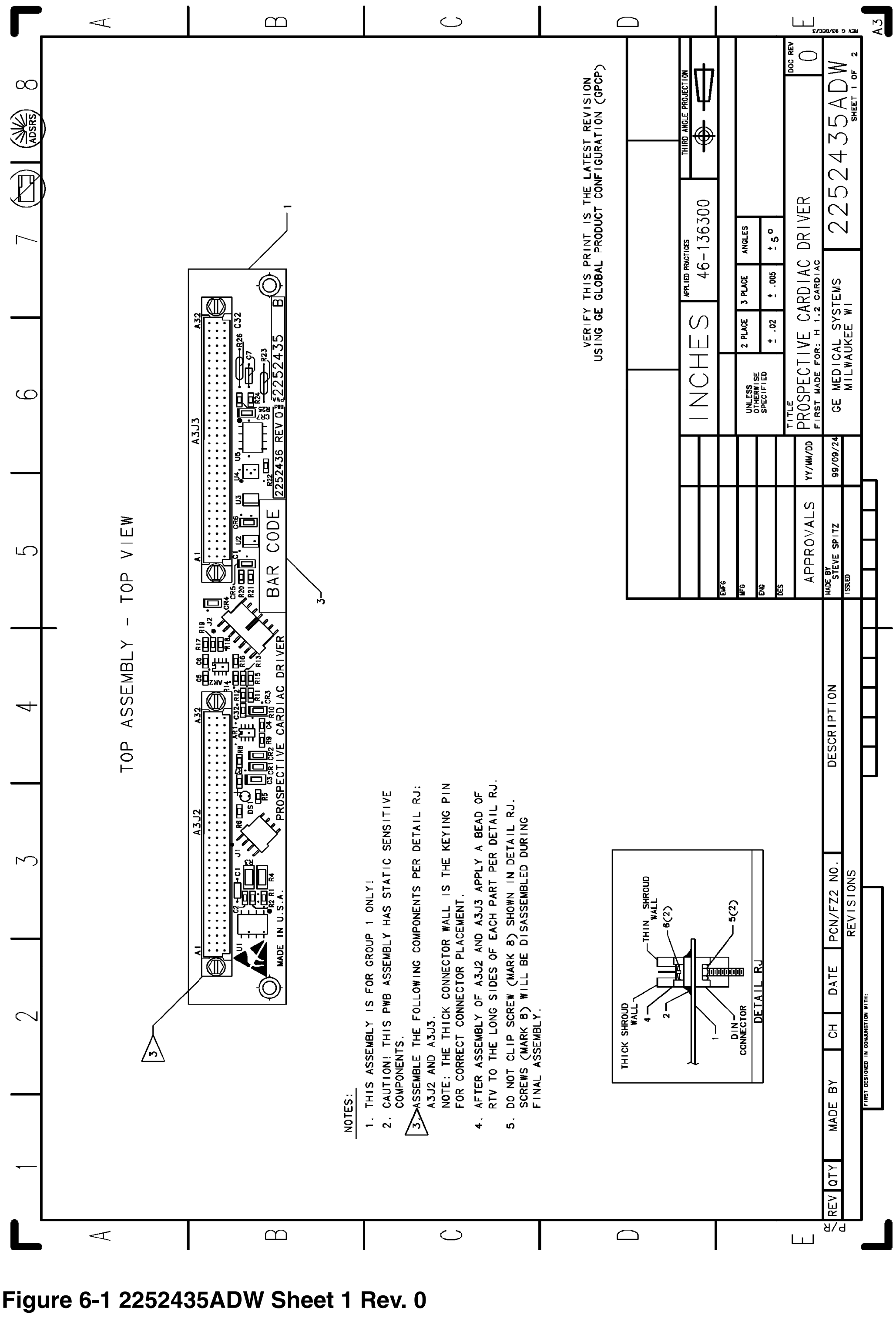


Figure 6-1 2252435ADW Sheet 1 Rev. 0

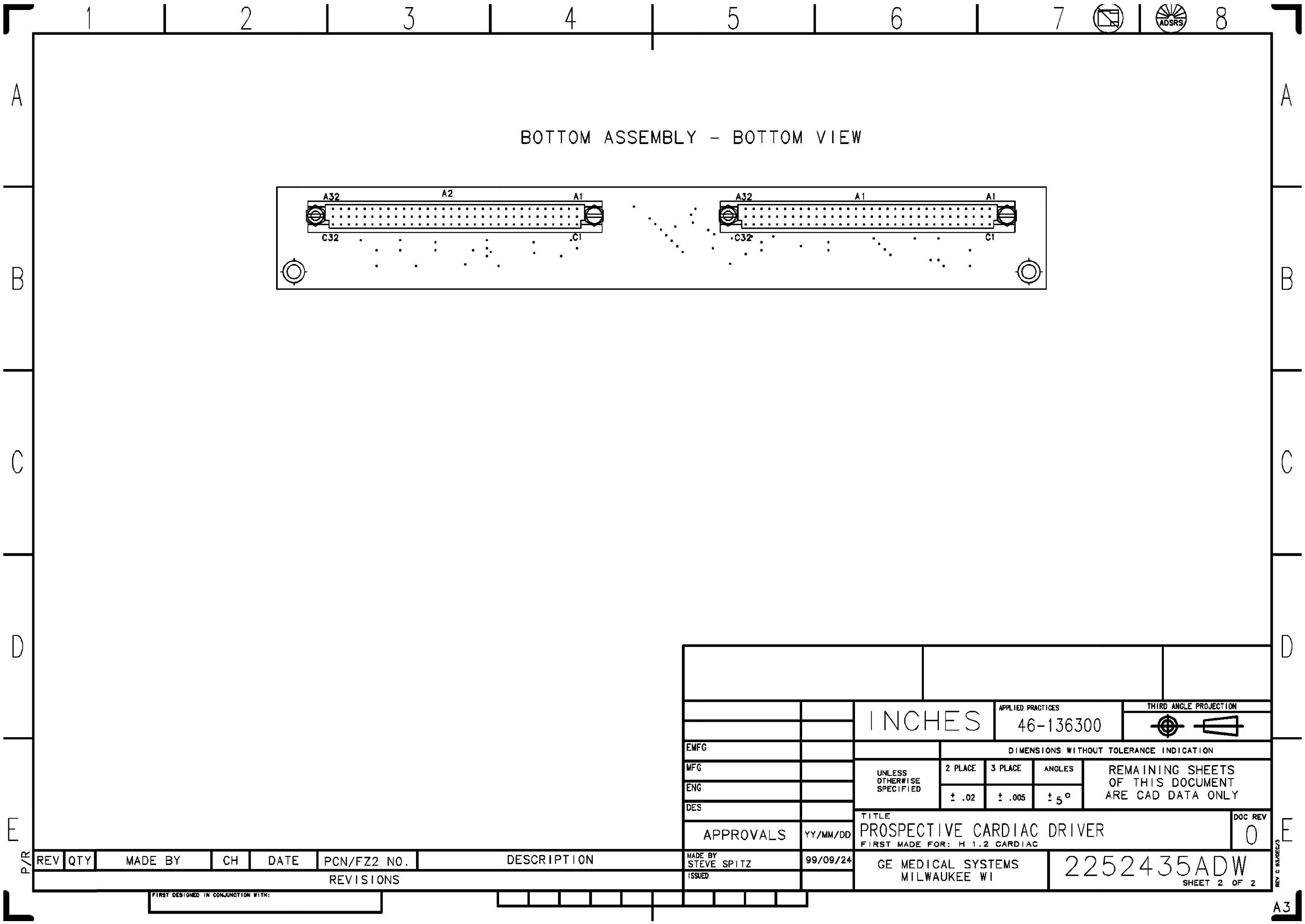


Figure 6-2 2252435ADW Sheet 2 Rev. 0

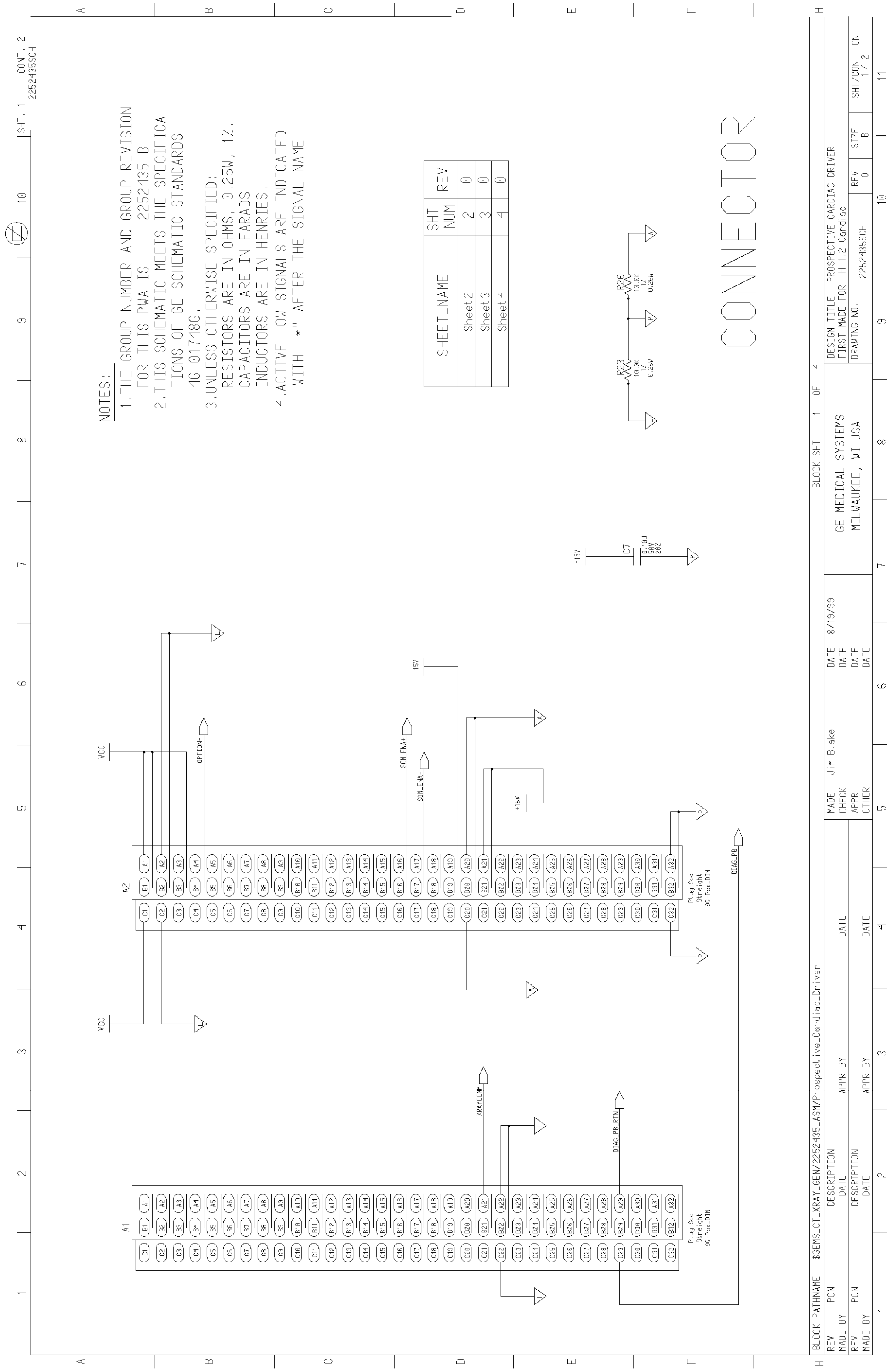


Figure 6-3 2252435SCH Sheet 1 Rev. 0

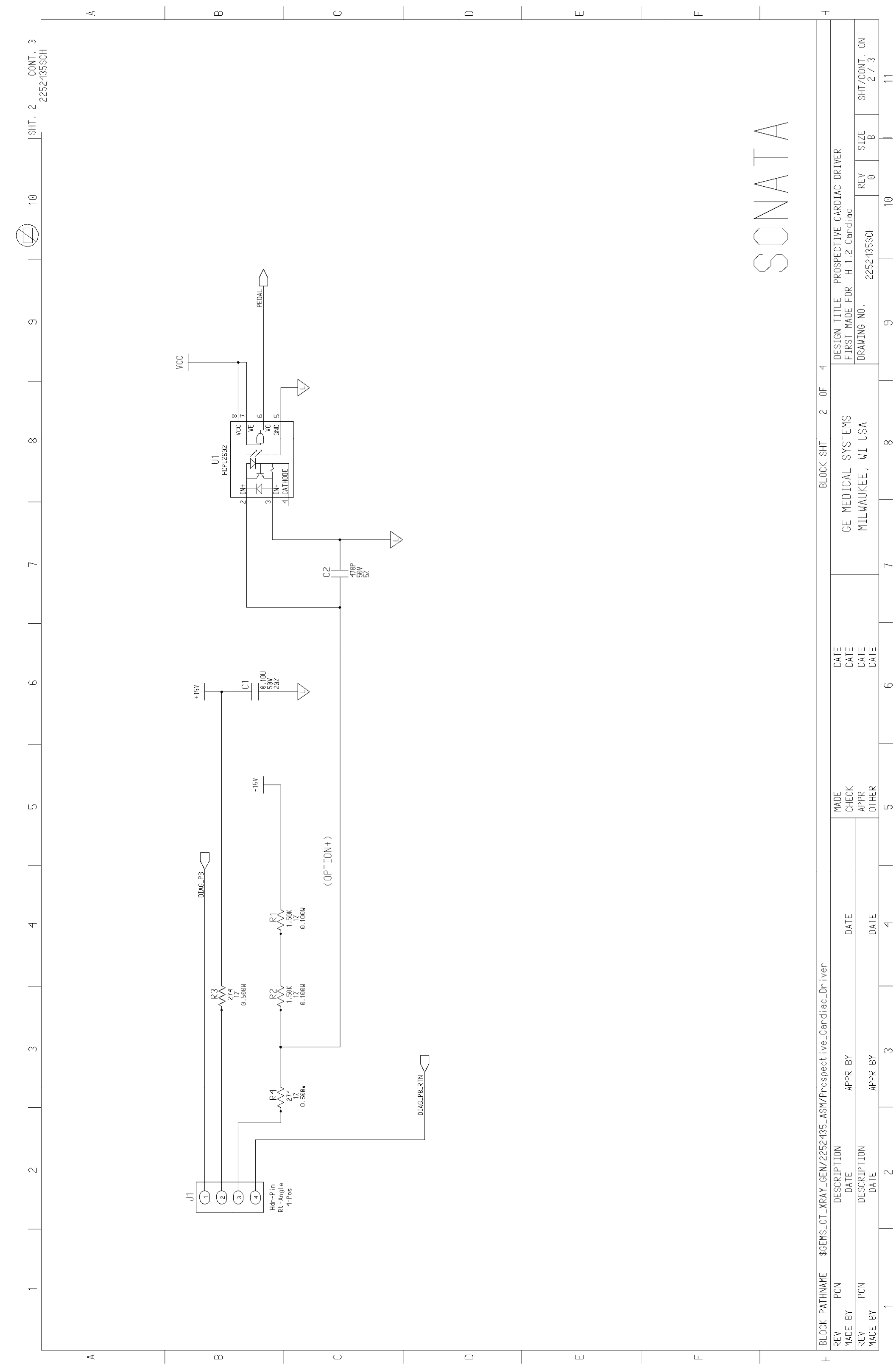


Figure 6-4 2252435SCH Sheet 2 Rev. 0

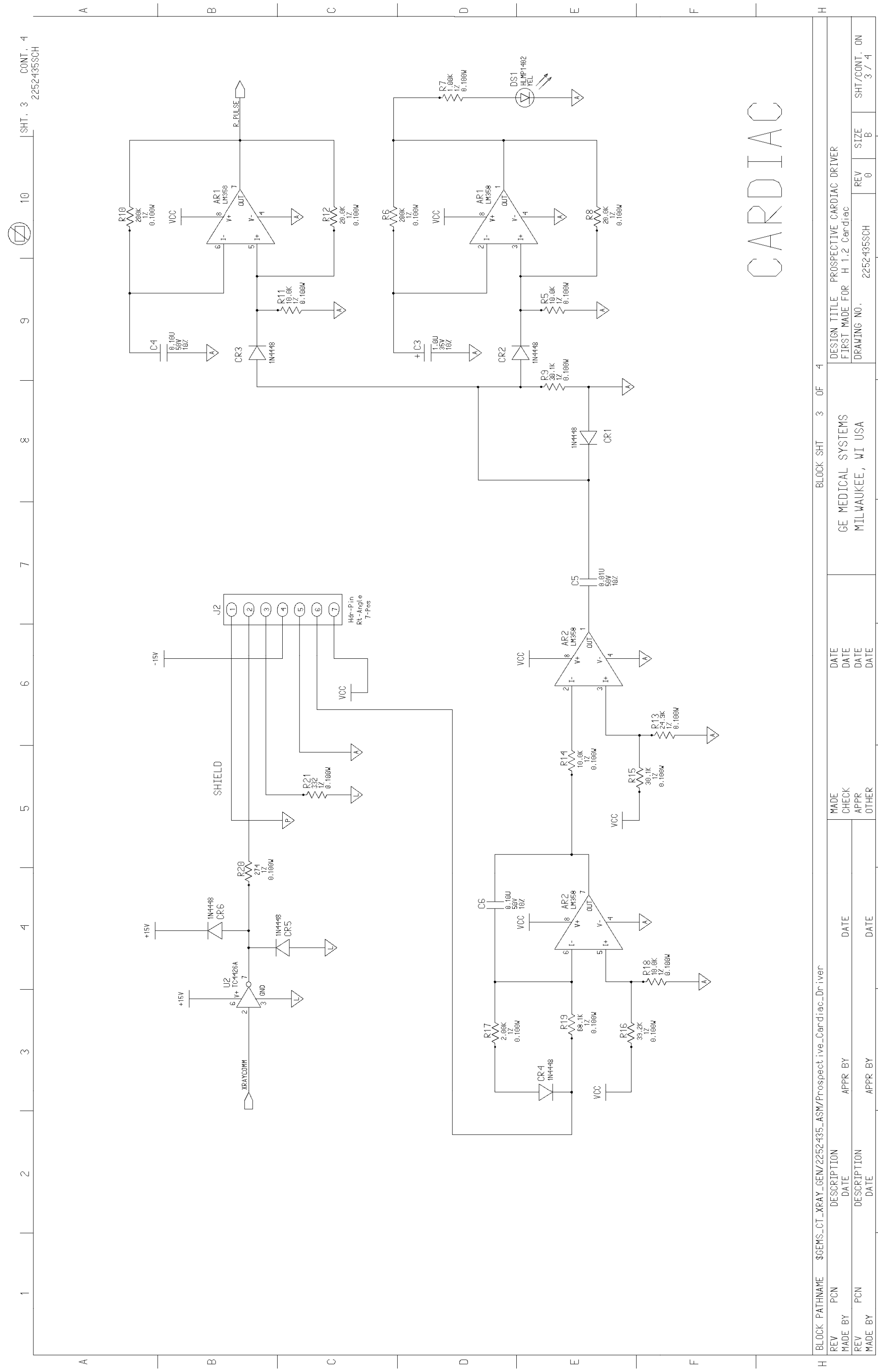


Figure 6-5 2252435SCH Sheet 3 Rev. 0

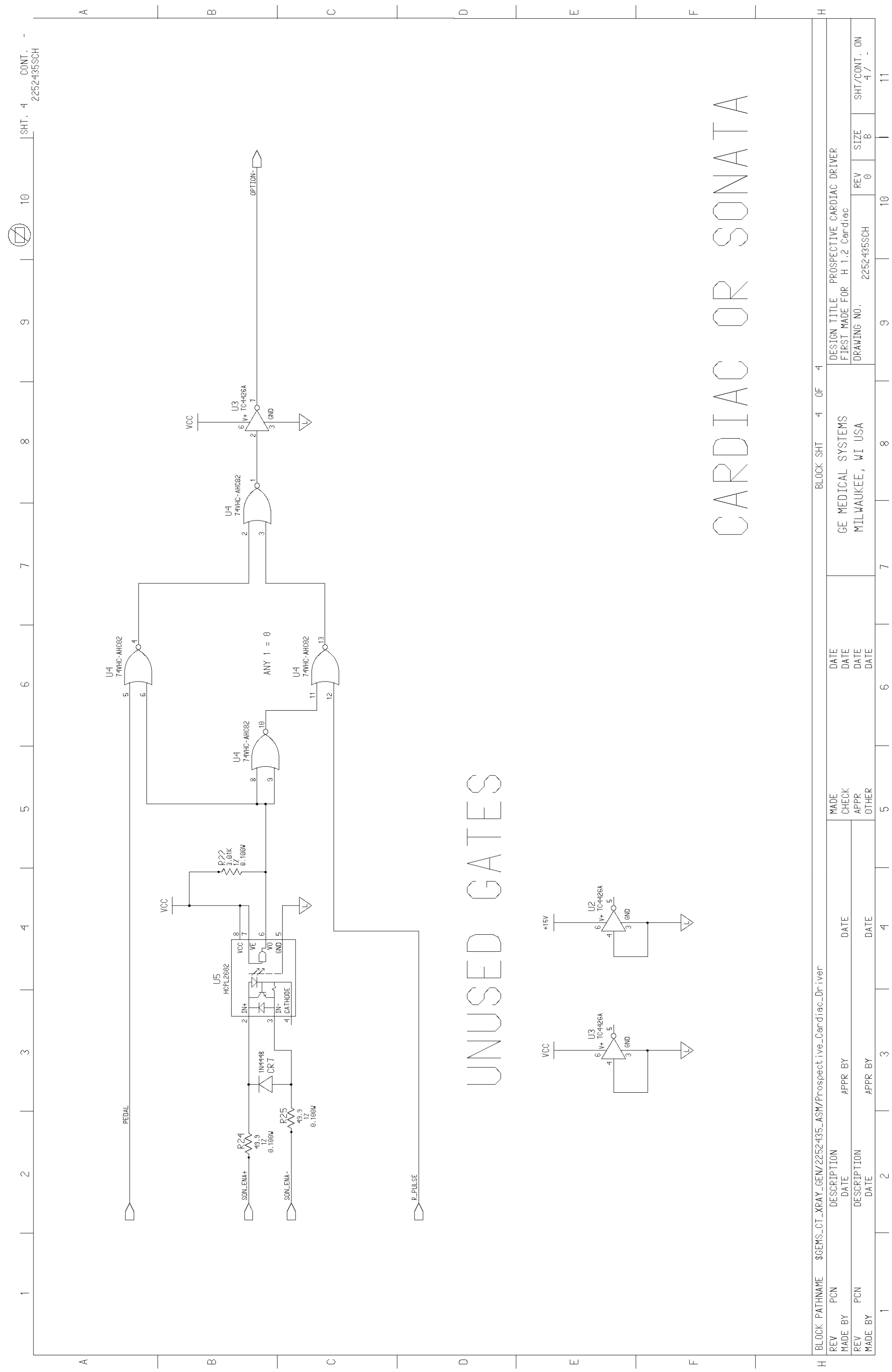


Figure 6-6 2252435SCH Sheet 4 Rev. 0

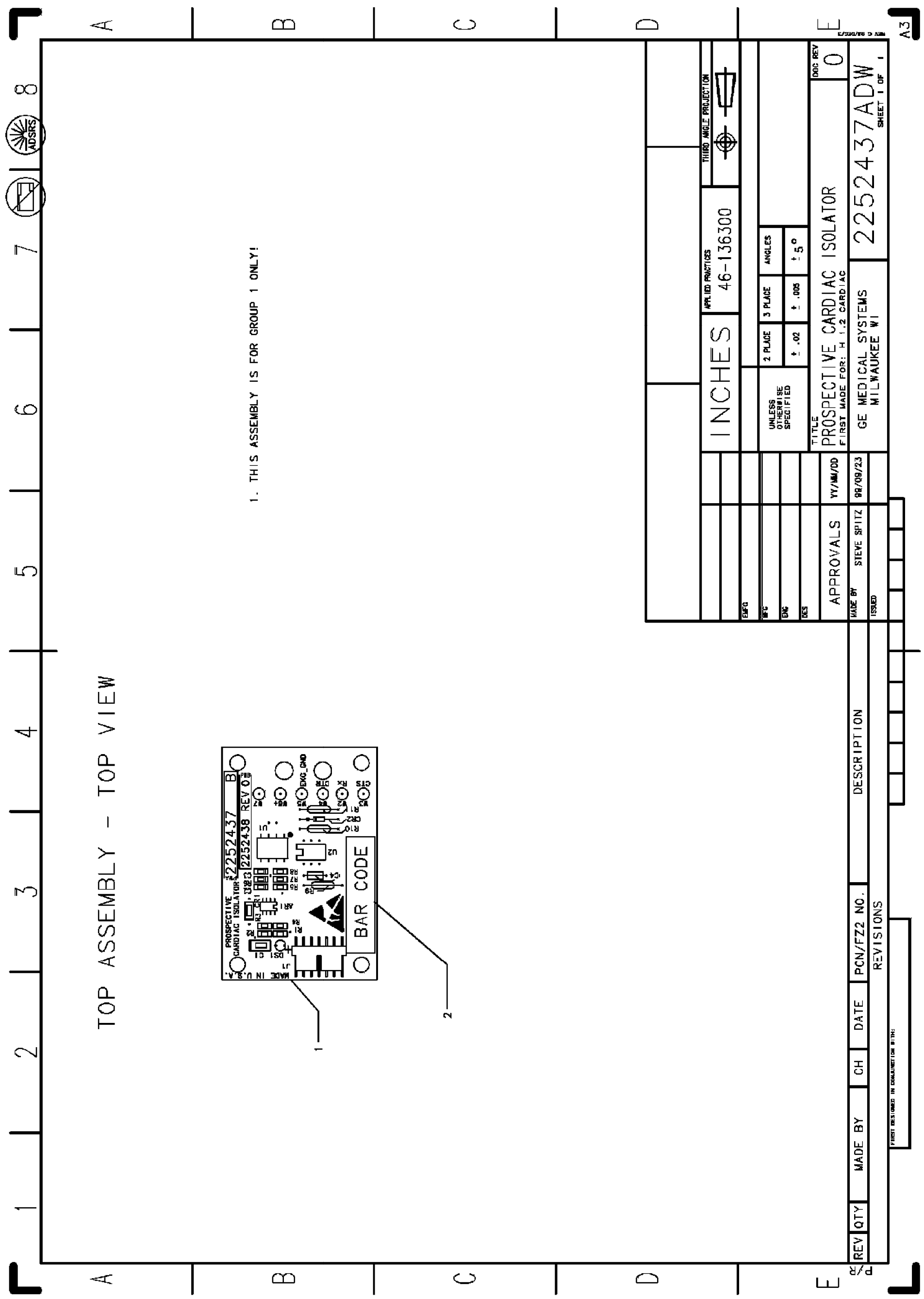


Figure 6-7 2252437ADW Sheet 1 Rev. 0

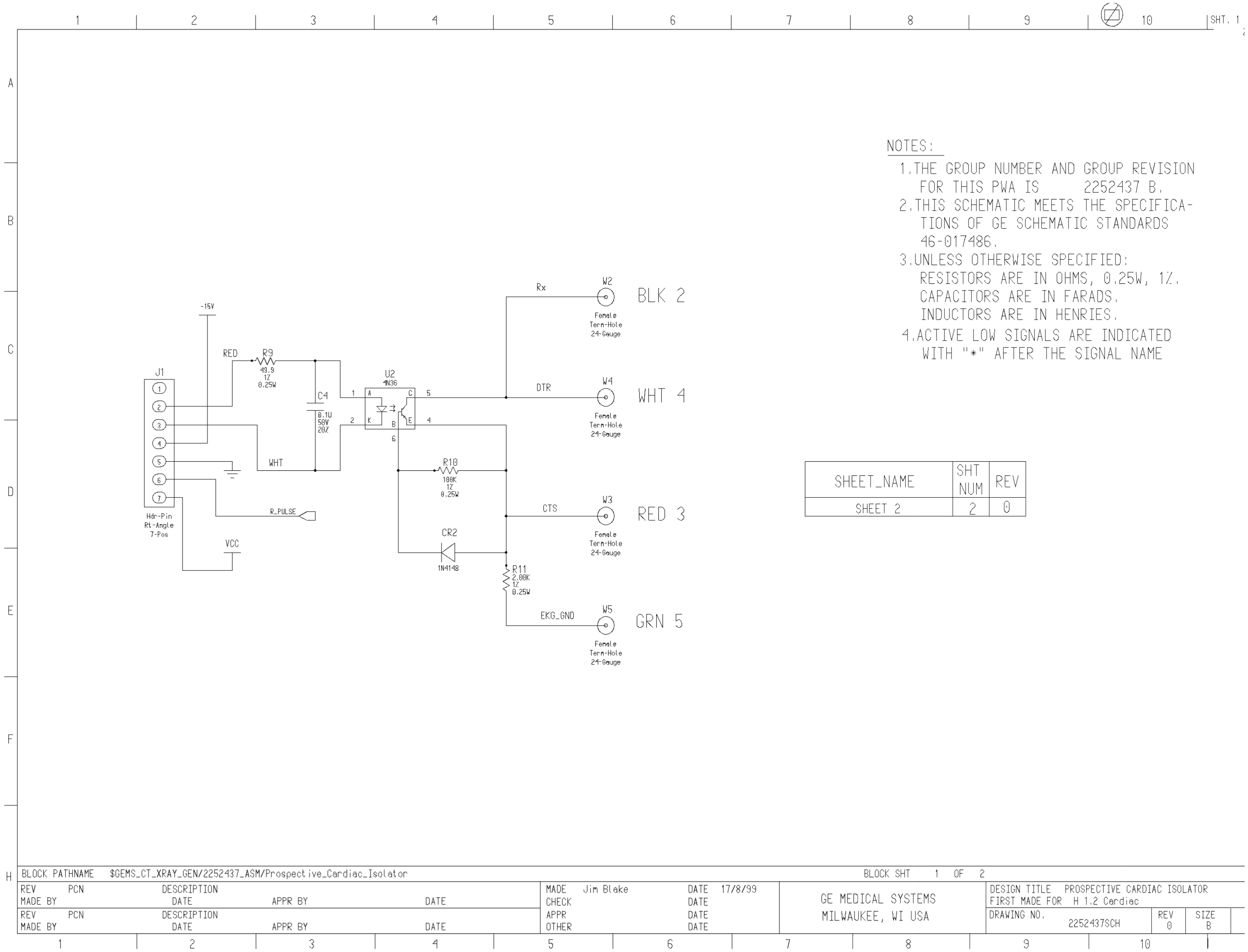


Figure 6-8 2252437SCH Sheet 1 Rev. 0

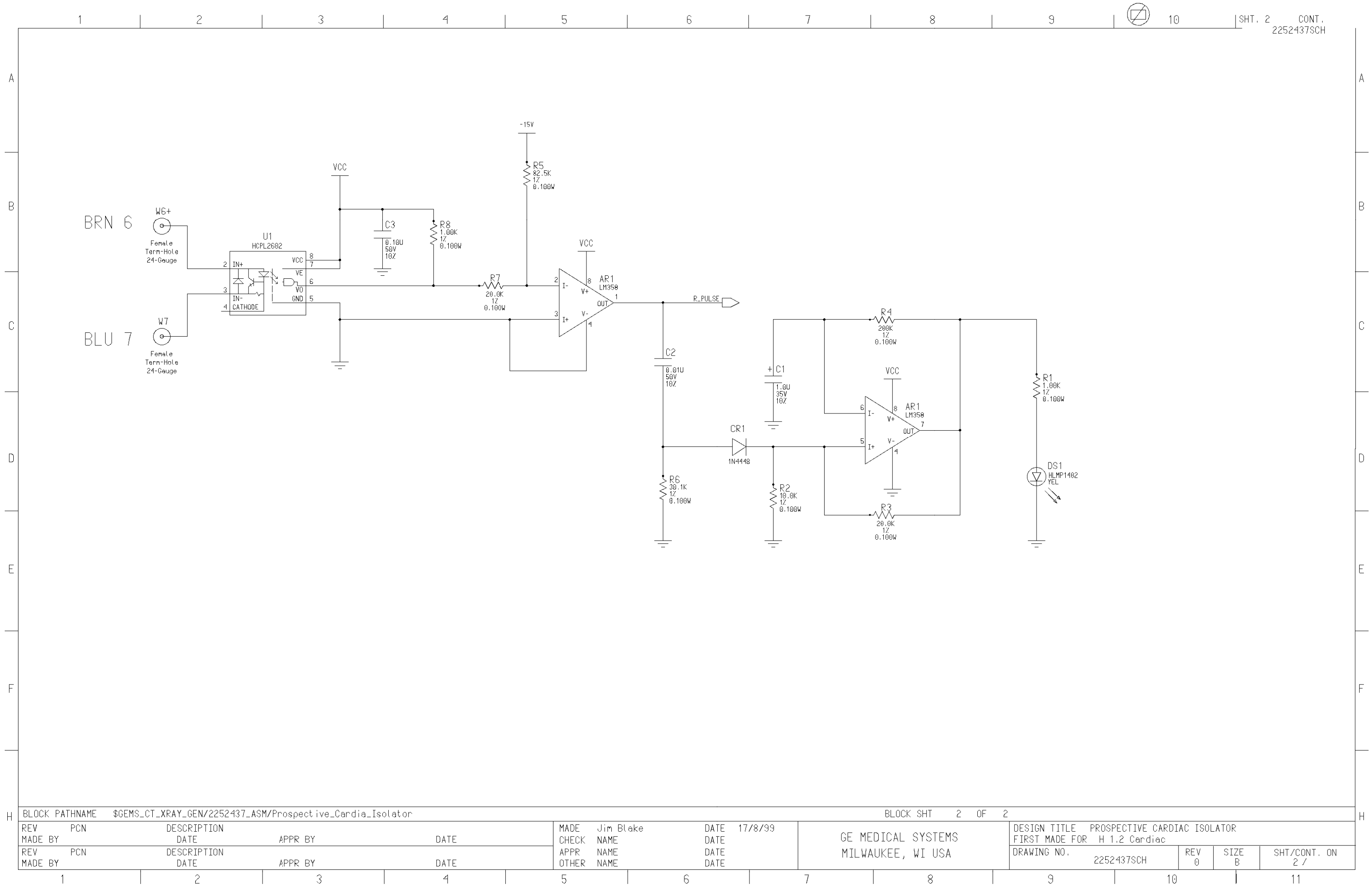
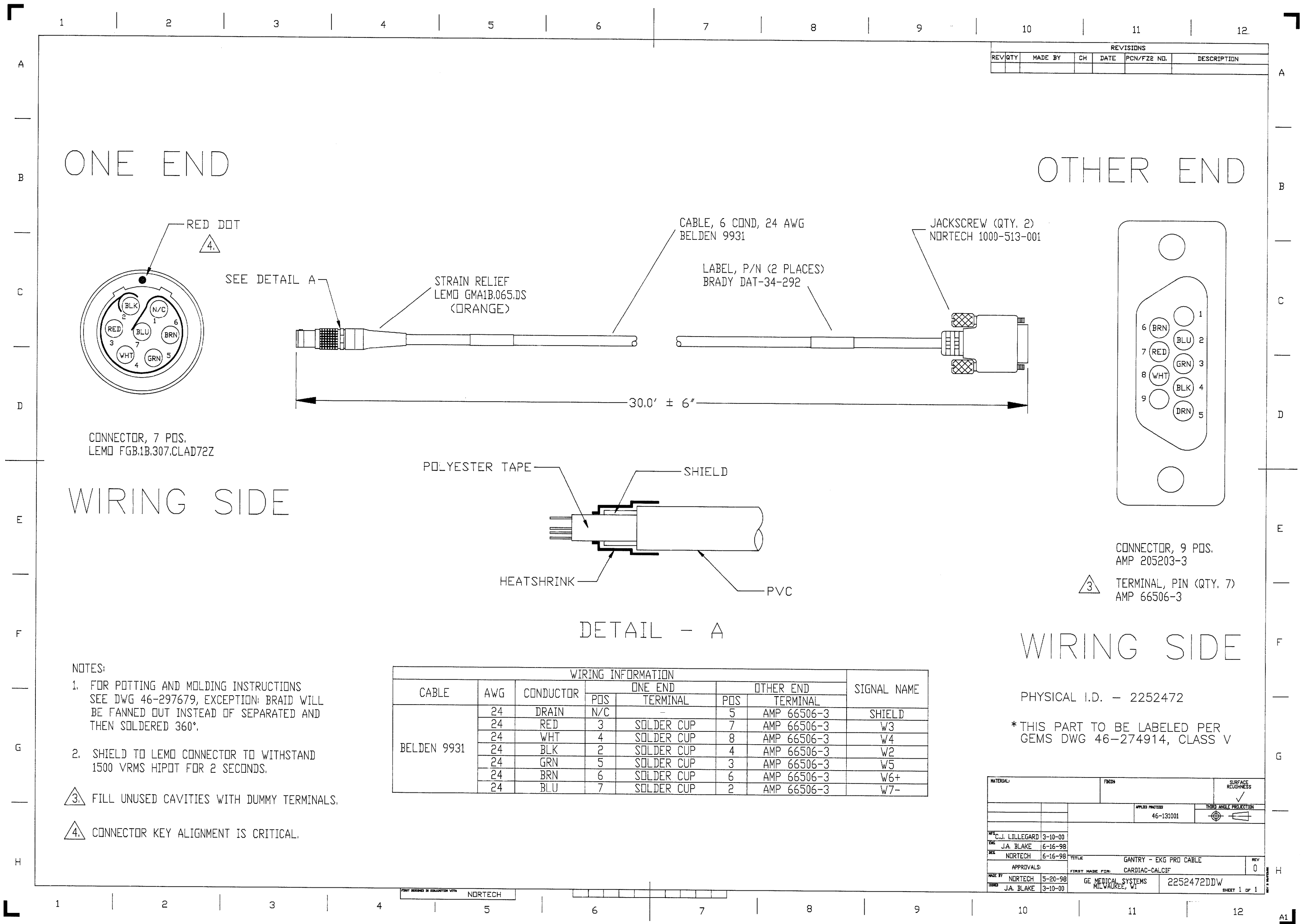


Figure 6-9 2252437SCH Sheet 2 Rev. 0



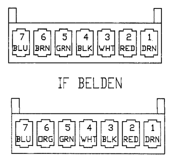
WIRING INFORMATION							
CABLE	AWG	CONDUCTOR	ONE END		OTHER END		SIGNAL NAME
			POS	TERMINAL	POS	TERMINAL	
BELDEN 9931	24	DRAIN	N/C	-	5	AMP 66506-3	SHIELD
	24	RED	3	SOLDER CUP	7	AMP 66506-3	W3
	24	WHT	4	SOLDER CUP	8	AMP 66506-3	W4
	24	BLK	2	SOLDER CUP	4	AMP 66506-3	W2
	24	GRN	5	SOLDER CUP	3	AMP 66506-3	W5
	24	BRN	6	SOLDER CUP	6	AMP 66506-3	W6+
	24	BLU	7	SOLDER CUP	2	AMP 66506-3	W7-

Figure 6-10 2252472DDW Sheet 1 Rev. 0

REVISIONS

REV	QTY	MADE BY	CH	DATE	PCN/FZ2 NO.	DESCRIPTION

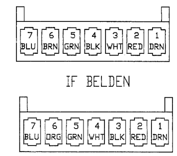
ONE END



CONNECTOR, 7 POS.
MOLEX 22-01-3077

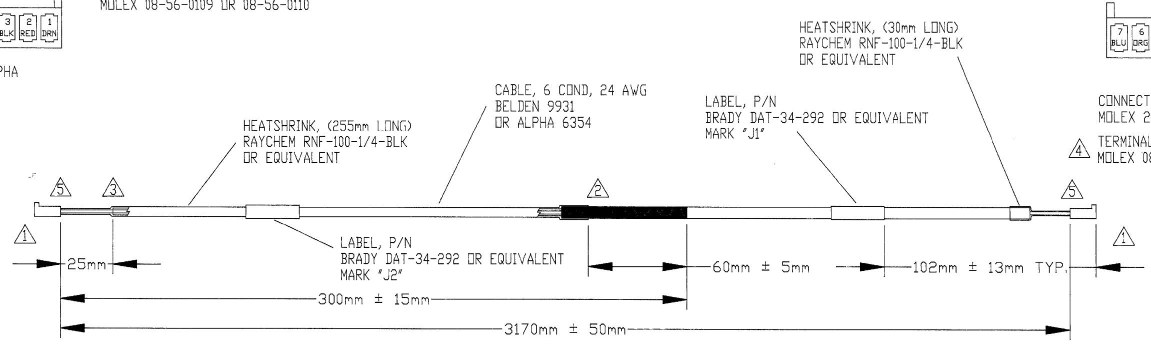
TERMINAL, WIPER (QTY. 7)
MOLEX 08-56-0109 OR 08-56-0110

OTHER END



CONNECTOR, 7 POS.
MOLEX 22-01-3077

⚠4 TERMINAL, WIPER (QTY. 7)
MOLEX 08-56-0109 OR 08-56-0110



WIRING SIDE

WIRING INFORMATION

CABLE	AWG	CONDUCTOR	ONE END POS	ONE END TERMINAL	OTHER END POS	OTHER END TERMINAL
BELDEN 9931	24	DRAIN	1	MOLEX 08-56-0109 OR 08-56-0110	1	MOLEX 08-56-0109 OR 08-56-0110
	24	RED	2	MOLEX 08-56-0109 OR 08-56-0110	2	MOLEX 08-56-0109 OR 08-56-0110
	24	WHT	3	MOLEX 08-56-0109 OR 08-56-0110	3	MOLEX 08-56-0109 OR 08-56-0110
	24	BLK	4	MOLEX 08-56-0109 OR 08-56-0110	4	MOLEX 08-56-0109 OR 08-56-0110
	24	GRN	5	MOLEX 08-56-0109 OR 08-56-0110	5	MOLEX 08-56-0109 OR 08-56-0110
	24	BRN	6	MOLEX 08-56-0109 OR 08-56-0110	6	MOLEX 08-56-0109 OR 08-56-0110
	24	BLU	7	MOLEX 08-56-0109 OR 08-56-0110	7	MOLEX 08-56-0109 OR 08-56-0110

WIRING SIDE

WIRING INFORMATION

CABLE	AWG	CONDUCTOR	ONE END POS	ONE END TERMINAL	OTHER END POS	OTHER END TERMINAL
ALPHA 6354	24	DRAIN	1	MOLEX 08-56-0109 OR 08-56-0110	1	MOLEX 08-56-0109 OR 08-56-0110
	24	RED	2	MOLEX 08-56-0109 OR 08-56-0110	2	MOLEX 08-56-0109 OR 08-56-0110
	24	BLK	3	MOLEX 08-56-0109 OR 08-56-0110	3	MOLEX 08-56-0109 OR 08-56-0110
	24	WHT	4	MOLEX 08-56-0109 OR 08-56-0110	4	MOLEX 08-56-0109 OR 08-56-0110
	24	GRN	5	MOLEX 08-56-0109 OR 08-56-0110	5	MOLEX 08-56-0109 OR 08-56-0110
	24	ORG	6	MOLEX 08-56-0109 OR 08-56-0110	6	MOLEX 08-56-0109 OR 08-56-0110
	24	BLU	7	MOLEX 08-56-0109 OR 08-56-0110	7	MOLEX 08-56-0109 OR 08-56-0110

PHYSICAL I.D. – 2252473

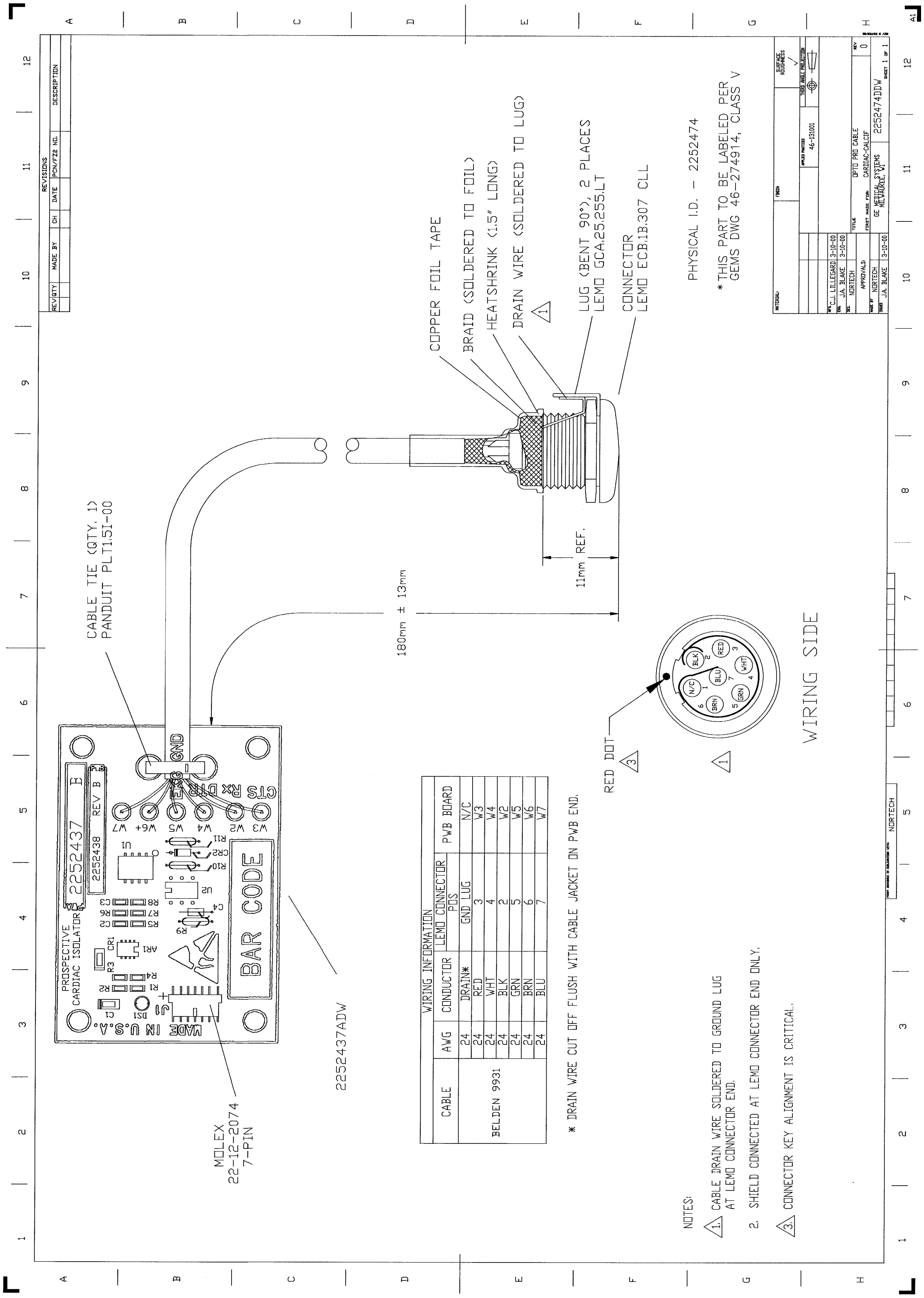
*THIS PART TO BE LABELED PER
GEMS DWG 46–274914, CLASS V

NOTE:

1. COVER DRAIN WIRE ON EACH END WITH HEATSHRINK (RAYCHEM RNF-100-1/16-BLK), 19mm LONG EACH.
2. OUTER BRAID IS EXPOSED 60mm, 1/4" HEATSHRINK TO OVERLAP BRAID 13mm.
3. END OF FOIL IS 25mm FROM CONNECTOR.
4. ALTERNATE MOLEX TERMINAL 08-56-0110 MAY BE USED.
5. HEATSHRINK IS FLUSH TO BACK SIDE OF CONNECTOR.

MATERIAL | FINISH | SURFACE ROUGHNESS
APPLIED PRACTICES 46-131001 | THIRD ANGLE PROJECTION
C.J. LILLEGARD 3-10-00
J.A. BLAKE 3-10-00
NORTECH
APPROVALS
TITLE DRIVER – ISOLATOR PRO CABLE
FIRST MADE FOR CARDIAC-CALCIF
REV 0
MADE BY NORTECH
J.A. BLAKE 3-10-00
GE MEDICAL SYSTEMS MILWAUKEE, WI.
2252473DDW
SHEET 1 OF 1
NORTECH

Figure 6-11 2252473DDW Sheet 1 Rev. 0



WIRING INFORMATION

CABLE	AWG	CONDUCTOR	LEMO CONNECTOR POS	PWB BOARD
BELDEN 9931	24	DRAIN*	GND LUG	N/C
	24	RED	3	W3
	24	WHT	4	W4
	24	BLK	2	W2
	24	GRN	5	W5
	24	BRN	6	W6
	24	BLU	7	W7

Figure 6-12 2252474DDW Sheet 1 Rev. 0

Chapter 7
Troubleshooting

Testing without EKG: *Please refer to Gantry to EKG Cable Schematic Figure 6-10 on page 48.*

The interface integrity can be evaluated using an ohm-meter when the EKG is not attached to the cable "D" connector at the monitor. X-Ray ON will be low when not scanning and will be high when scanning.

1.) With all cables properly connected and an EKG simulator attached to the EKG monitor, LED DS1 will flash with each trigger pulse signal. This is true on both the Opto Isolator Board and the Driver Board.
2.) Make sure the Gantry to EKG Cable (2252472) is connected to the gantry base.
3.) Disconnect the Gantry to EKG Cable from the EKG monitor.
4.) Measure the voltages at the connector end attached to the monitor.

The following ohm-meter readings would be expected at the "D" connector in a normal installation:

Condition	Connection			Measured Value
(SIGNAL)	(Positive)	To	(Negative)	
X-RAY ON Low	Pin 5	to	Gantry Ground	open
X-RAY ON Low	Pin 3	to	Gantry Ground	open
X-RAY ON Low	Pin 4	to	Pin 8	short
X-RAY ON Low	Pin 3	to	Pin 7	2 k ohms
X-RAY ON Low	Pin 8	to	Pin 7	< 3 k ohms
X-RAY ON High	Pin 5	to	Gantry Ground	open
X-RAY ON High	Pin 3	to	Gantry Ground	open
X-RAY ON High	Pin 4	to	Pin 8	short
X-RAY ON High	Pin 3	to	Pin 7	2 k ohms
X-RAY ON High	Pin 8	to	Pin 7	> 20 k ohm

Table 7-1 Nominal Values - On cable (2252472) at Monitor Connection Side

The Invivo 3500CTP EKG machine is serviced by Invivo.
For Service or Questions please call 1-800-331-3220 or 407-275-3220

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006